

A CONGRUENCE BETWEEN MULTINOMIAL COEFFICIENTS

by P. DELIGNE

Institut des Hautes Études Scientifiques, Bures-sur-Yvette

Let p be a prime number. For every nonzero integer m , let $v(m)$ denote the exponent of the highest power of p dividing m . We also put $v(0) = \infty$.

For a k -tuple $\mathbf{n} = (n_1, \dots, n_k)$ of integers ≥ 0 , set

$$|\mathbf{n}| = \sum_1^k n_i, \quad a\mathbf{n} = (an_1, \dots, an_k), \quad X^{\mathbf{n}} = \prod_1^k X_i^{n_i}, \quad \mathbf{n}! = \prod_1^k n_i!.$$

Let $C(\mathbf{n})$ be the coefficient of the monomial $X^{\mathbf{n}}$ in

$$\left(\sum_1^k X_i \right)^{|\mathbf{n}|};$$

these are the multinomial coefficients. Thus

$$(1) \quad \left(\sum_1^k X_i \right)^n = \sum_{|\mathbf{n}|=n} C(\mathbf{n}) X^{\mathbf{n}}$$

$$(2) \quad C(\mathbf{n}) = \frac{|\mathbf{n}|!}{\mathbf{n}!}.$$

For $k = 2$, these are the binomial coefficients:

$$(3) \quad C(n_1, n_2) = \binom{n_1 + n_2}{n_1}.$$

Proposition 1. *Let $\mathbf{n}$ be a k -tuple of integers ≥ 0 , and put $n = |\mathbf{n}|$. If p is a prime number ≥ 5 , then*

$$(4) \quad C(p^l \mathbf{n}) \equiv C(\mathbf{n}) \pmod{p^{v(n)+3}}.$$

EXAMPLE. — For $p = 5$, $k = 2$, $l = 1$, and $\mathbf{n} = (1, 1)$, one has

$$\binom{10}{5} = 252 = \binom{2}{1} + 2 \cdot 5^3.$$

PROOF. — It is known, or follows from (2), that

$$(5) \quad \left(\sum_1^k X_i \right)^{p^l} = \left(\sum_1^k X_i^{p^l} \right) + pQ,$$

where Q is a polynomial with integer coefficients, of degree $< p^l$ in each variable. The binomial formula then gives

$$(6) \quad \left(\sum_1^k X_i \right)^{p^l n} = \left(\sum_1^k X_i^{p^l} \right)^n + \sum_1^n p^i \binom{n}{i} R_i,$$

$$(7) \quad R_i = \left(\sum_1^k X_i^{p^l} \right)^{n-i} Q^i.$$

For two polynomials A and B with integer coefficients, write $A \equiv B \pmod{r}$ if the coefficients of $X^{p^l n}$ in A and B are congruent modulo r . With this notation, (4) is rewritten as

$$(8) \quad \left(\sum_1^k X_i \right)^{p^l n} \equiv \left(\sum_1^k X_i^{p^l} \right)^n \pmod{p^{v(n)+3}}.$$

Lemma 1.

$$(9) \quad v\left(\binom{n}{i}\right) \geq v(n) - v(i).$$

Let the additive group $\mathbb{Z}/(n)$ act on itself by translation. Thus each $x \in \mathbb{Z}/(n)$ determines the permutation $y \mapsto x + y$ of $\mathbb{Z}/(n)$. If a subset P with i elements in $\mathbb{Z}/(n)$ has stabilizer H , that is, if $H \cdot P = P$, then P is a union of cosets of H , and $\#H$ divides i . The cardinal

$$r_P = n/\#H$$

of the orbit of P under $\mathbb{Z}/(n)$, acting on the power set $\mathcal{P}(\mathbb{Z}/(n))$, therefore satisfies

$$p^{v(n)-v(i)} \mid r_P.$$

Since the set of subsets of $\mathbb{Z}/(n)$ with i elements is a disjoint union of orbits, it follows that

$$p^{v(n)-v(i)} \mid \binom{n}{i}.$$

Lemma 2. For $p \neq 2, 3$ and $i \geq 3$, one has

$$(10) \quad p^i \binom{n}{i} \equiv 0 \pmod{p^{v(n)+3}}.$$

Indeed,

$$v\left(p^i \binom{n}{i}\right) = i + v\left(\binom{n}{i}\right) \geq v(n) + (i - v(i))$$

by Lemma 1. If $3 \leq i < p$, then $v(i) = 0$ and $i - v(i) \geq 3$. If $i \geq p$, then

$$i - v(i) \geq i - \log i / \log p.$$

The last function of i is increasing for $i \geq p$, and its value at $i = p$ is $p - 1 \geq 3$.

Lemma 3. *The coefficient of $X^{p^l n}$ in R_1 is zero.*

No monomial $X^{\mathfrak{r}}$ appearing in Q has an exponent $\mathfrak{r}$ divisible by p^l . The polynomial

$$R_1 = \left(\sum_1^k X_i^{p^l} \right)^{n-1} Q$$

inherits this property.

By Lemmas 2 and 3,

$$(11) \quad \left(\sum_1^k X_i \right)^{p^l n} \equiv \left(\sum_1^k X_i^{p^l} \right)^n + \binom{n}{2} p^2 R_2 \pmod{p^{v(n)+3}}.$$

This congruence already implies, by Lemma 1, that

$$(12) \quad C(p^l n) \equiv C(n) \pmod{p^{v(n)+2}}.$$

The only monomials $X^{\mathfrak{r}}$ with exponent divisible by p^l which occur in $(pQ)^2$ are the monomials

$$X_i^{p^l} X_j^{p^l} \quad (i < j).$$

The coefficient of each of them is

$$(13) \quad \sum_{i \neq 0, p^l} \binom{p^l}{i} \binom{p^l}{p^l - i} = \sum \binom{p^l}{i}^2 - 2 = \binom{2p^l}{p^l} - 2.$$

By (12),

$$\binom{2p^l}{p^l} - 2 \equiv \binom{2p}{p} - 2 \pmod{p^3}.$$

Thus

$$\left(\sum_1^k X_i \right)^{p^l n} \equiv \left(\sum_1^k X_i^{p^l} \right)^n + \binom{n}{2} \left[\binom{2p}{p} - 2 \right] R_2 \pmod{p^{v(n)+3}},$$

with R_2 a polynomial with integer coefficients.

Since

$$v\left(\binom{n}{2}\right) \geq v(n)$$

by Lemma 1, the proposition follows from the next lemma.

Lemma 4. *For every prime $p \geq 5$,*

$$\binom{2p}{p} \equiv 2 \pmod{p^3}.$$

For $1 \leq i \leq p-1$, one has

$$p^{-1} \binom{p}{i} \equiv (-1)^{i-1} i^{-1} \pmod{p}.$$

Hence

$$(14) \quad p^{-2} \left[\binom{2p}{p} - 2 \right] = \sum_1^{p-1} p^{-2} \binom{p}{i}^2 \equiv \sum_1^{p-1} (1/i)^2 \pmod{p}.$$

Define

$$z \in \mathbb{Z}/(p), \quad z = \sum_{i \in \mathbb{Z}/(p)^*} 1/i^2.$$

For $u \in \mathbb{Z}/(p)^*$, one has

$$z = \sum_{i \in \mathbb{Z}/(p)^*} 1/(u^{-1}i)^2 = u^2 z,$$

and therefore

$$(u^2 - 1)z = 0.$$

Since $p \neq 2, 3$, there exists u such that $u^2 \neq 1$, and hence $z = 0$.

Formula (14) consequently becomes

$$p^{-2} \left[\binom{2p}{p} - 2 \right] \equiv 0 \pmod{p},$$

which completes the proof.